



Cockroach milk

A thankfully short video showing the crystalline milk coming out of sacs in the stomach of the cockroach. <https://digit.in/jul20-74>



Cockroaches are cool

Watch the high speed videos and crushing forces applied on cockroaches to make the CRAM robot. <https://digit.in/jul20-75>

quantities. The protein crystals have an added benefit. They maintain an equilibrium with the solution around them, and as the solution is used up, through digestion, the protein crystals release more proteins at the rate at which the proteins are used up. This makes for a timely release of food that keeps providing nutrition on ingestion, at a steady rate. This property, along with the density of the packed proteins, lipids and sugars makes it one of the most nutritious substances known to scientists. The cockroach milk can be used to replace the milk in your morning cup of coffee, and it is going to taste mostly the same, only look different. The crystals remain stable when suspended in water. In the future, this cockroach milk could be available for mass consumption.

SEARCH AND RESCUE ROBOTS

You might have seen roaches scurry away and disappear into incredibly tiny gaps after you attempt to stamp them. This is because cockroaches can bear tremendous amounts of load without injuring themselves, and they can also move nearly as fast when they are flattened, by changing the orientation of their legs. It does not matter if a gap is 1 cm or 0.5 cm, the critter can move equally fast. The cockroaches can flatten their bodies all the way to 2 mm, and still manage to move freely. They can also withstand forces that are 900 times more than their own body weight.

Researchers from Berkeley inspired by these hardy and versatile roaches, decided to make robots for search and rescue operations with similar properties. This robot was dubbed the compressible robot with articulated mechanisms or CRAM. The robot is cheap to manufacture, is about the size of a human palm, and has a plastic covering on its back similar to the wings of cockroaches. When it has to navigate through tiny gaps, CRAM can spread out its legs similar to real cockroaches. The robot has been designed to navigate the rubble,

which is tricky for robots in the aftermath of a natural disaster such as an earthquake or a tornado. Swarms of similar robots can be used to explore the tricky debris, to try and identify survivors, or idea routes for rescue workers. To develop the innovative leg moments of the robots, the team of scientists used high-speed film of cockroaches running through narrow gaps in laboratory settings.

Another team from Berkeley were also inspired by the cockroach to create another robot prototype for search and rescue operations. The approach uses a piezoelectric material which causes the surface to expand or contract when a current is applied. A plastic coating over the piezoelectric material causes the material to bend instead of expanding and contracting. This property was used to create

and nerves. Creating prosthetic or robotic hands has been notoriously difficult because it has, so far, been impossible to give them the full range of capabilities of a human hand. The approach so far was to keep making the robots smarter, while keeping the basic design of the fingers the same. Harvard researchers came up with the innovative idea of redesigning the fingers from scratch, inspired by the legs of cockroaches. Such graspers allow robotic hands and prosthetics to improve their performance, even if they are not made much smarter than they already are. Humans use the sense of touch on their fingers to estimate relationships between their hands and the objects that they are trying to manipulate. Robots and prosthetics have to make do with sensors, that are sometimes not

located on the hand itself. These sensors make the graspers expensive, and more complex, at the same time. The flexible, springy legs of the cockroaches provided the necessary design inspiration here. These legs can navigate uneven surfaces with ease,



The CRAM robot inspired by cockroaches

a robot that leaps frogs ahead when current is supplied in bursts. The robot has a single leg, and can dash about on the floor at speeds comparable to cockroaches, or twenty times its own body length every second. The robot weighs less than one tenth of a gram, but continues to function even after 60 kgs of pressure is applied on it. This robot design is tethered to a wire at the moment, which supplies the necessary current for movement, but the team is working on making a wireless version.

PROSTHETICS

The hand is one of the most complex parts of the human body, with a number of bones supporting an intricate networks of tendons, muscles, veins

because of their very structure, and do not need the tiny cockroaches to think. Indeed the cockroaches do not have too many brain cells to spare for such complicated thinking. By redesigning the fingers to incorporate the jaggy joints on a cockroach leg, the Harvard researchers managed to create a versatile robotic hand that could easily manipulate a wide range of objects with ease, from tennis balls, to eggs to wine glasses. The design had an additional advantage of reducing the weight of the prosthetic arms, which is always good. In its present state though, the cockroach-inspired robotic arm cannot manipulate very tiny objects such as keys and forks, which is something that the researchers are working on. **4**